

CIS 106

Problems – Using Nested if and Compound Relational Conditions

For each problem, develop the IPO and Code.

The input to the problem is quantity of widgets. Your program should determine the price to charge based on the schedule below. Calculate the extended price (quantity x price). Calculate tax at 7%. Display the extended price, tax amount and total.

Quantity	Price
>10000	\$10
5000 to 10000	\$20
Below 5000	\$30

INPUT	PROCESS	OUTPUT
Quantity of widgets	If the quantity_of_widgets is ≥ 10000 , the rate is \$10 Else, if the quantity of widgets is 5000 to 10000, the rate is \$20 Else, the rate is \$30	Display the extended price, tax amount, and total.
	Extended_price = quantity_of_widgets x price Tax = extended_price x 0.07%	

Enter a part number and quantity Determine the cost per unit using the table below. Then calculate the total cost (quantity x unit cost). Display the part number, cost per unit and total cost. Note: Part number can be an integer but it can also be a string because you are not doing arithmetic on it. However in your code if statement be sure to compare using consistency, that is, if item == "10" when item is a string and if item == 10 when item is an integer.

Part	Unit Cost
10 or 55	1.00
99	2.00
80 or 70	3.00
All others	5.00

INPUT	PROCESS	OUTPUT
Part number	If part_number == '10' or '55' then unit_price is \$1.00 Else if part_number == '99' then unit_price is \$2.00 Else if part_number == '80' or '70' then unit_price is \$3.00 Else unit_price is \$5.00	Display the part number, cost per unit and total cost.
Quantity	total_cost is quantity x unit_price	

Enter a principle amount of a CD and year to maturity of CD. Determine the interest rate based on the amount of the principle **and** maturity (see below). Calculate first year interest (principle x interest rate). Display principle, interest rate and the interest amount for first year.

Principle	Years to Maturity	Interest Rate
>\$100,000	5	6%
\$50,000 to \$100,000	10	5%
\$50,000 to \$100,000	5	4%
Any other principle and years		2%

INPUT	PROCESS	OUTPUT
Principle amount of a CD	If principal_amount is >=\$100,000 and years_of_maturity == 5 interest_rate is 6% Else if principle_amount >=\$50,000 and years_of_maturity ==10 interest_rate is 5% Else if principle_amount >=\$50,000 and	Display principle, interest rate and the interest amount for first year.

	years_of_maturity ==10 interest_rate is 4% Else Interest_rate is 2%	
Year of maturity of CD	First_year_interest= principle_amount x interest_rate	

Allow the user to enter number of concert tickets. The price per ticket depends on the volume (see below). Display the number of tickets, price per ticket and the total cost (number of tickets x Price Per Ticket).

Quantity	Price Per Ticket
>=25	\$50
10 to 24	\$60
5 to 9	\$70
Less 5	\$75

INPUT	PROCESS	OUTPUT
Enter the number of concert tickets	If ticket_number is >=25 price_per_ticket is \$50 Else if ticket_number is 10 to 24 price_per_ticket is \$60 Else if ticket_number is 5 to 9 price_per_ticket is \$70 Else price_per_ticket is \$75	Display the number of tickets, price per ticket and the total cost (number of tickets x Price Per Ticket).
	total_cost= ticket_number x price_per_ticket	

The user will enter employee last name, salary and job level (as noted below). Use the job level to determine the bonus rate. Then compute bonus to be salary times bonus rate. Display employee last name and bonus.

Job Level	Bonus Rate
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10 and above	25%
5 to 9	20%
All others	10%

INPUT	PROCESS	OUTPUT
Employee's last name	If job_level is >==10 bonus_rate is 25% Else if job_level is 5 to 9 bonus_rate is 20% Else bonus_rate is 10%	Display the employee's last name and bonus.
Salary	Bonus = bonus_rate x job_level	
Job Level		